

Quick Activity 1: School Supplies Calculator

A notebook costs ₱45, and a pen costs ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the **total cost** and **change**.

Assume that quantities are positive whole numbers and the payment is enough to cover the purchase.

INPUT	PROCESS	OUTPUT
Quantity of notebooks, quantity of pens, payment amount	Total cost = quantity of notebooks + quantity of pens Change = payment - total cost	Total cost, change

DECOMPOSITION:

- Get the number of notebooks
- Get the number of pens
- Get the payment amount
- Calculate the number of notebooks
- Calculate the number of pens
- Calculate the cost
- Calculate the change
- Get the total cost
- Get the change

PSEUDOCODE:

START

```
READ notebooks, pens, payment
notebookSubtotal ← notebooks x 45
penSubtotal ← pens x 12
totalCost ← notebookSubtotal + penSubtotal
Change ← payment - totalCost
DISPLAY totalCost, change
```

END

DRY-RUN CALCULATION: QUICK ACTIVITY ONE

CASE TEST ONE:

A notebook costs **₱45**, and a pen costs **₱12**. A student buys several notebooks and pens and pays **₱200**.

STEP	VARIABLE	CALCULATION	RESULT
1	notebookSubtotal	3×45	₱135.00
2	penSubtotal	2×12	₱24.00
3	totalCost	$135 + 24$	₱159.00
4	change	$200 - 159$	₱41.00

EXPECTED OUTPUT:

Total cost: ₱159.00

Change: ₱41.00

CASE TEST TWO:

A notebook costs **₱45**, and a pen costs **₱12**. A student buys several notebooks and pens and pays **₱150**.

STEP	VARIABLE	CALCULATION	RESULT
1	notebookSubtotal	2×45	₱90.00
2	penSubtotal	3×12	₱36.00
3	totalCost	$90 + 36$	₱126.00
4	change	$150 - 126$	₱24.00

EXPECTED OUTPUT:

Total cost: ₱126.00

Change: ₱24.00